

Auditory and Olfactory Stimuli-Based Attractors to Induce Reorientation in Virtual Reality Forward Redirected Walking

Jieun Lee

School of Integrated Technology, Gwangju Institute of Science and Technology (GIST)
jieunlee@gm.gist.ac.kr

Kyunghwan Kim

School of Integrated Technology, Gwangju Institute of Science and Technology (GIST)
kyunghwankim@gm.gist.ac.kr

Seokhyun Hwang

School of Integrated Technology, Gwangju Institute of Science and Technology (GIST)
anoldhsh@gist.ac.kr

Seungjun Kim*

School of Integrated Technology, Gwangju Institute of Science and Technology (GIST)
seungjun@gist.ac.kr

ABSTRACT

Redirected walking (RDW) visually manipulates the virtual environment to imperceptibly redirect the walkers to keep them in the tracking area, and offers a larger space than physical space. An attractor is a redirected walking technique that captures the walker's attention and manipulates the walker's trajectory through rotational gain. However, the attractor visually manipulates the walker's virtual environment using a predefined rotational gain, and having to constantly gaze at the attractor or the attractor frequently appearing until the walker's direction matches the desired direction, are problems limiting the application of visual attractors. Moreover, when the walker is unable to recognize or ignores the attractor, reorientation fails. In this study, we designed a human-sense-stimulating attractor that utilizes the auditory and olfactory senses to improve the rotational gain, naturalness, and immersion and decrease the chance of reorientation failure. Although sound and scents are invisible, they can be detected through direction; however, humans cannot recognize the accurate direction of a sound or scent. Based on these characteristics, auditory and olfactory attractors are proposed. We measured the amount of reorientation induced by the auditory and olfactory attractors and calculated the reorientation success rate. Additionally, the naturalness and immersion of the attractor were evaluated. The auditory attractor has a high reorientation success rate, naturalness, and immersion. The olfactory attractor induces more turn changes in the walker than other attractors, and a high number of turn changes leads to a larger rotational gain. Auditory and olfactory-based attractors have the potential to overcome the shortcomings of visual attractors such as the frequent interventions.

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519719>

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms.

KEYWORDS

Virtual Reality (VR), Redirected walking, Attractor, Auditory, Olfactory

ACM Reference Format:

Jieun Lee, Seokhyun Hwang, Kyunghwan Kim, and Seungjun Kim. 2022. Auditory and Olfactory Stimuli-Based Attractors to Induce Reorientation in Virtual Reality Forward Redirected Walking. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519719>

1 INTRODUCTION

In virtual reality (VR), walking is a natural body movement that leads to a high sense of being physically present in the virtual space. Virtual locomotion methods such as walking in place, arm swings, and walking by cycling utilize body movements to support VR walker, allowing them to experience the feelings of natural walking and being present in the space [9, 18, 37]. Despite allowing walkers to travel in the virtual world, current walking methods mimicking locomotion are insufficient to offer a real walking experience. Real walking is an irreplaceable locomotion that allows for a higher sense of presence in VR [12, 38]. Ideally, walkers can walk everywhere in the virtual world while physically remaining and walking in albeit confined space. However, the most challenging aspect of real walking in VR is that walkers must remain within a maximum area of 5×5 m, which is referred to as the tracking area of the commercial VR system [14]. Only when the walker is within this tracking area does the VR system offer the user interfaces or contents in the correct locations, based on the specific location and viewing direction of the walker. This system also ensures the walker's safety by preventing crashing or falling, by setting a virtual boundary, and warning not to cross it. Walker movement of the walker is constrained by the limited walkable area and the VR system frequently interrupts the walker to ensure they remain in the tracking area.

Therefore, redirected walking (RDW) is proposed to allow walkers to walk in a virtual environment (VE) that is larger than the real environment [29, 34]. RDW manipulates the VE and creates an

intentional mismatch between the physical and perceptual movements of the walker. Owing to visual dominance, the walker believes the distorted VE, and loses a sense of direction, thereby unknowingly moving to arrive at the desired position [10]. For example, RDW can distort the visual scene displayed through a VR head-mounted display, such that walkers perceive that they are walking straight, when they are actually walking in different directions, such as a curve. In other instances, RDW increases or decreases the degree of rotation in the virtual world as the user rotates, causing the user to ultimately stand in a direction that is different from the physically rotated one.

An attractor is a well-known rotational RDW technique that captures the visual attention of the walker and imperceptibly manipulates the rotational degree of the virtual world with the predefined rotational gain. The attractor calls for physical engagement of the walker, for example by looking at or interacting with the attractor [32]. The walker becomes inattentive blind, and often cannot detect the rotation of the VE, resulting in the walker naturally recognizing the rotated virtual world as the original one [35]. The attractor first stops the walker, and then manipulates the virtual world; therefore, this method is effective in terms of safety. However, the visual attractor frequently intervenes to change the direction of the walker until the walker rotates in the desired direction, which can easily disrupt the virtual presence by appearing in the field of vision [24, 30]. The frequent appearance and the small reorientation by predefined rotational gain decrease the effectiveness of the reorientation. In addition, the attractor utilizes primarily visual stimuli; therefore, walker may fail to notice it, potentially causing them to escape the tracking area or become stuck in a corner.

To improve the limitations of visual manipulation of attractor, there are several types of research regarding the integration of other sensory modalities with a visual attractor, and suggestions for other sensory modality-based attractors to increase the efficiency of the attractor. Audio-based attractors have been suggested to examine the effectiveness of audio stimulation-based manipulations, and the results indicate that such methods can enhance redirection effectiveness [7, 31]. Scent has also been used as a supplemental element to increase immersion and emotional pleasure; it is less obvious as an attractor because scent can be diluted and dispersed in the air [6, 27]. Sound and scent are invisible and express spatial information, and these senses are less sensitive than vision [17]. Compared with human vision, auditory or olfactory perception is less effective for us to locate the source of the corresponding stimuli or to pinpoint the direction in which we should be reoriented; however, they can guide individuals in specific situations, such as detecting stimuli accruing behind them and in the dark [10, 25]. These properties could effectively reorient the walker in the VE by increasing their rotation number which increases the rotational gain. In addition to visual attractors, different types of attractors that stimulate human sensation need to be studied to expand human sensation applications and address the limitations of current visual-based attractors more effectively to reorient the walker.

Therefore, we propose three different types of attractors that can stimulate to the human visual, auditory, and olfactory senses, which can capture walker attention and induce frequent reorientation. These attractors consist of two components: a reorientation

trigger and visual feedback. Figure 1 shows an example of the different attractors for reorientation in the VE, where the reorientation trigger of each auditory and olfactory stimulating attractor is the corresponding auditory and olfactory stimuli. The visual feedback in this process is the visual reward combined with the reorientation trigger. The visual attractor uses the visual object as the reorientation trigger and another visual object as visual feedback, while in multi-attractor, the reorientation trigger is an integrated auditory and olfactory stimulus, and visual feedback is the reward. We investigated the possibility of using auditory and olfactory stimulating attractors in the role of reorientation by measuring the number of walker turn changes and comparing the reorientation success rate. The naturalness and immersion of each attractor were evaluated by the walkers. Moreover, we interviewed the participants on how to increase the effectiveness of auditory and olfactory attractors compared to that of visual-only attractors.

2 RELATED WORKS

This section discusses RDW techniques in the virtual world, suggested applications of the attractors, and the use of other human senses in stimulating redirection.

2.1 Redirected walking

Decision-making of humans is mainly influenced by visual information, that is, the walker recognizes the visual scene as a result of their body movement and change in direction [10, 26]. The RDW technique uses this phenomenon to keep the walker in the tracking area by offering a walkable space larger than the real space. RDW is a software-based technique that distorts the VE without the walker noticing the manipulation and redirects the walker [29]. In the first proposed RDW experiment, the VE was rotated according to the walker's movement, who walks in a zigzag pattern in the virtual world but actually moves back and forth in the given room. Since then, diverse redirection methods have been attempted to manipulate the VE and redirect the walker in the desired direction. Through translation, rotation, and curvature-related RDW techniques, the walkers walked the VE without noticing the manipulation and felt a larger space than the real space [3, 33]. The bending curvature RDW method, which is a different curvature method, was proposed to compact the required space for curvature RDW and apply it on a room-scale [15]. Another RDW technique is based on the walker's saccade eye movement. When the eye movement is detected, the visual scene is slightly distorted, and the walker walks around twice as wide as the actual space without noticing the manipulation [36]. Many RDW techniques subtly use visual manipulation imperceptibly, so there is a risk of redirection failure when the walker believes that their behavior is not manipulated.

2.2 Attractor

The attractor appears in front of the walker and captures their visual attention, after which, they are reoriented to the desired location with the predefined rotational gain for safety and proper content offering. Capturing the visual attention decreases the possibility of redirection failure [20]. Diverse types of attractors were studied when designing the attractor that captures the walker's attention for a long time. The attractor with a realistic appearance and sound



Figure 1: Schematic of human sensation stimuli-based attractors. (a) VR walker walks freely and approaches the boundary of the tracking area, and (b) visual, olfactory, auditory, or a combination of auditory and olfactory attractors induce the reorientation of the walker.

was preferred to other types, such as low-fidelity attractors and those with audio instructions [22]. The content- and interactivity-relevant attractors were designed to increase reality and attention time. The interactivity-relevant attractor involved watching and clicking, and the result confirmed that a higher complexity of interaction captures the walker's attention for a longer period [4, 5]. Moreover, the attractor asked the walker to perform visuomotor synchronized activities, which decreased dizziness and increased the presence, while the walkers were unaware of the VE manipulation [32]. However, the attractor mainly uses visual effects and manipulation, so it has limitations in terms of visual manipulation. The walker needs to continuously focus on and follow the attractor until their direction matches the desired direction. This method helps users explicitly figure out the direction to walk in. However, the process is time-consuming and exhausting. The attractor frequently appears and interrupts the walker when they walk near

the boundary and the corners, and then it easily reduces the presence. To decrease the frequency of appearance of the attractor, the walker's trajectory is narrowed to a given path, and they cannot freely travel the VE.

2.3 Auditory attractors and olfactory sense stimulation

Other sensory modalities, such as the auditory and olfactory senses, assist in increasing the immersion in VR and act as attractors for reorientation. When only auditory stimulation worked as the attractor, the results confirmed visual dominance; however, it showed the possibility that an auditory attractor can present spatial information that helps the user to reorient, especially when the visual situation is unclear or complex [7, 31]. Sound can create contextual effects suitable for a given VR situation (e.g., birdsong in the forest,

horn sounds in the city) and provide spatial information, while causing much less visual interference than visual attractor. Scent was barely used as an attractor for user reorientation in VE; however, it is very effective in changing mood and enhancing the emotional effect [1, 27, 28]. The olfactory feedback in VR also provides more immersive VEs and increases the feeling of the presence. As scents easily disperse and remain in the air, a mixture of diverse scents creates confusion, and an air conditioning system is required for ventilation. However, auditory and olfactory stimulation can direct even if they are invisible [25]. This property could overcome the visual attractor's limitations, such as continued visual engagement. Humans' hearing sensation detects the source of a sound somewhat accurately, but is not perfect. Additionally, it is difficult for humans to find the exact origin of the scent. This property has the potential to be an effective attractor that can induce observable behavior and a larger angle of walker reorientation [17]. Indeed, these stimulations can induce the user to rotate more than visual stimulation alone, and the rotational gain increases because it is accumulated according to the user's rotational movement [11, 23]. The large reorientation angle and the small number of interventions of the attractor might increase the effectiveness of the attractor. In this study, we designed human sense-stimulating attractors consisting of the reorientation trigger and visual feedback parts, and observed their use as attractors for improving the limitations of the visual attractor.

3 SYSTEM DESIGN AND EXPERIMENT

3.1 VE and attractors

The VR system used in this research supports a maximum of 5 m × 5 m of tracking space. We designed the virtual world to look like a forest using Unity, wherein a tent and bonfire were used to represent a campsite, and a dog sat near the tent. The background and bonfire produced crickets' chirping sound and firewood burning sound, respectively, to increase immersion and hide real-world noise. We also set the red virtual wall grid to withdraw the participants from the wall. When a participant neared the wall, we used the attractor to capture the user's attention, induced reorientation, and spawned flowers, positioned diagonally backward at 45° and 2 m away from the look-at direction of the user, as visual feedback (see Figure 1(b)).

The three attractors used visual, auditory, olfactory, and multiple auditory and olfactory methods to stimulate the user's senses. The experimenter used a VR controller to make a flying bird appear as a visual attractor in front of the user's view and capture the user's attention [22] (see Figure 2(a)). The auditory and olfactory attractors, the dog's whines and spraying of floral fragrance, respectively, worked behind the user [21], who could hear the auditory stimuli through the VR head-mounted display (HMD) and smell the olfactory stimuli from the perfume spraying system, as shown in Figures 2(b) and (c). The floral fragrance spraying system has two ultrasonic humidifier modules controlled by an Arduino nano board and Bluetooth signal from Unity, and each outlet is on both sides of the VR HMD to indicate the left and right back directions [1]. The movement of a participant wearing the VR HMD integrated with the scent spraying system without the additional ventilation system might dilute the sprayed fragrance quickly. For the multi-attractor, auditory and olfactory stimuli were activated simultaneously.

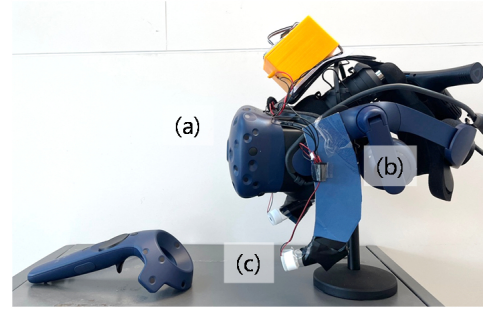


Figure 2: Visual, auditory, and olfactory stimulating attractors through the VR HMD; a visual attractor displayed in VR HMD (a), an auditory attractor sounds from the headset (b), and an olfactory- and multi-attractors diffuses perfume via ultrasound humidifiers (c).

3.2 Experiment procedure

We recruited 20 participants (6 female, 14 male; age [$M = 24.85$, $SD = 1.90$]) with no VR sickness. Before wearing the VR HMD, they were instructed to use their sight, hearing, and olfactory senses to find a flower and stop the experiment if they felt sick at any time. Initially, the participants walked freely around the virtual forest to find the flower, although the flower did not exist in the virtual forest. If the participants neared the virtual wall and needed a change in direction, a flower would appear behind the walker, on the left or right (see Figure 1). Simultaneously, the visual attractor appeared in front of the participants, and the dog as a part of the auditory- and multi-attractors appeared in the same place as the flower. The floral scent was sprayed from behind, as a part of the olfactory- and multi-attractors. All the attractors directed the participants to find the flower. Only one attractor appeared at a time, and each experiment was conducted twice in a randomized order, following the Latin square. If the participants ignored or did not recognize the attractors and crossed the boundary, we stopped the experiment and recorded that the reorientation failed. When the participants found and clicked on the flower, they were asked three questions—how they found the flower, how heavily did the attractor affect their reorientation, and any comment about the attractor which they experience at that time. In each experiment, we also recorded the participants' reorientation success and the number of times the walker's turn changed from the moment the attractor appeared until the participant located the flower. We measured the head-turning direction change, from left to right or right to left to count the number of direction changes. The reorientation success rate implied that the attractor surely catches the attention of the walker and induced reorientation toward the desired direction, and a high number of turn direction changes implied a larger reorientation angle that could decrease the number of the attractor appearance and intervention. After completing the experiments, the participants responded to a 5-point Likert scale questionnaire about the naturalness of the attractors (1 = not at all, 5 = strongly natural) and ranked them in order of the level of immersion of four different attractor types. Additionally, we inquired about the use

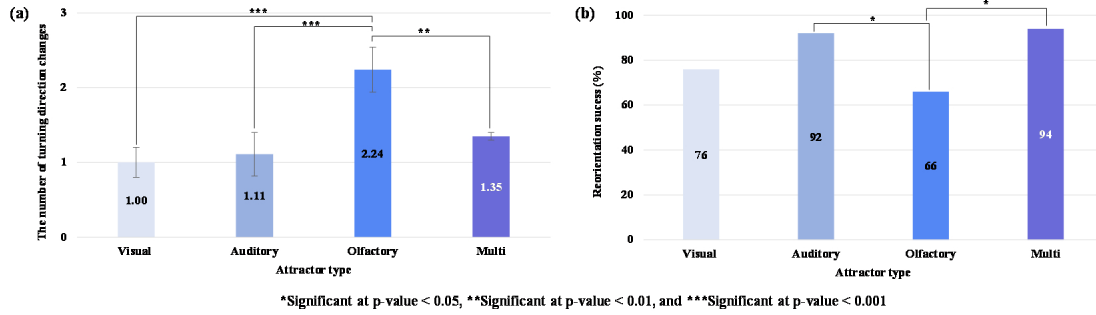


Figure 3: Participants' reorientation results: (a) the number of turning direction changes and (b) reorientation success rate.

of auditory and olfactory attractors to increase their effectiveness and applications in VR.

4 RESULTS AND DISCUSSION

We compared the participants' behavior such as the number of turning direction changes and reorientation success rate according to the various attractor types as well as the naturalness and immersion of the attractors. A Kruskal-Wallis test showed that the attractor affected the number of turning direction changes ($H(3) = 33.72, p = 0.000$). The visual attractor ($M = 1.00, SD = 0.00$), auditory attractor ($M = 1.11, SD = 0.29$), and multi-attractor ($M = 1.35, SD = 1.17$) had similar values, whereas the olfactory attractor ($M = 2.24, SD = 1.83$) induce more turn changes than other the attractors, as shown in Figure 3 (a). Post-hoc tests of the four attractors were conducted using Bonferroni-adjusted alpha levels of 0.0125 per test (0.05/4). The olfactory senses of humans cannot distinguish an accurate direction of the smell, and olfaction stimulation affects ambiguous visual situations [13, 17]. In our experiment situation that the participants attempted to find the flower but the flower exists behind the walker, so the olfactory attractor affects participants' behavior. The high number of turn direction changes in the olfactory attractor has the potential to increase the angle of VE manipulation by accumulating the angle according to the walker's turn, decreasing the intervention of the attractor. In the case of the visual attractor, once participants gazed at the visual attractor, they did not divert their eyes until they located the flower. Because the visual attractor requires the visual attention of the walker, participants continually stare at it. The auditory and multi-attractors had a lower number of rotation direction changes than the olfactory attractor, indicating that participants recognized the approximate direction of the sound stimuli, finding the flower without turn changes. In particular, the frequent rotations and greater rotational gains have advantages in scenarios that supported free walking without requiring precise rotations, such as travel and item-finding tasks.

The reorientation success rate of each attractor is shown in Figure 3 (b). Despite having a visual reorientation trigger and participants constantly gazing at the visual attractor, the visual attractor's success rate was not high. Some participants recognized the visual attractor as an element of the VE; a realistic and content-relevant attractor enhanced the presence, which could impede the reorientation work [32]. Others could not gaze at the visual attractor because the visual attractor had latency when the walkers' turning

was rapid. The olfactory attractor had the lowest reorientation success rate. Personal differences in perceptual sensitivities were the main cause of differences in the success rate of reorientation, and some participants also commented on the difficulty of smelling the floral scent because of an issue with their olfaction sensor [8, 19]. Second, the floral scent is a commercial floral perfume, whereas some participants considered the olfactory stimulation to be an air freshener. Auditory and multi-attractors have a high success rate. Although they do not have visual reorientation triggers, they can reorient walkers well. Additionally, the sound stimulus of the multi-attractor predominantly affects recognition and reorientation; therefore, the success rate of the multi-attractor is similar to that of the auditory attractor.

After participants completed the tasks, they were asked about the naturalness and immersion of the four types of attractors. We conducted the Kruskal-Wallis test ($H(3) = 24.19, p = 0.000$), and all attractors scored above three points of naturalness. The most natural attractor was the auditory attractor ($M = 4.60, SD = 0.50$), while the olfactory attractor ($M = 3.10, SD = 1.02$) had the lowest naturalness; however, the other attractors had a similar, low naturalness (visual attractor: $M = 3.45, SD = 0.89$; and multi-attractor: $M = 3.75, SD = 1.02$) (Figure 4 (a)). Participants ranked the attractors in order of immersion; a small number indicates a higher immersion rank, and a large number indicates a lower immersion rank (Figure 4 (b)). The auditory attractor ($M = 1.65, SD = 0.99$) was the most immersive, followed by the multi-attractor ($M = 2.20, SD = 0.95$), olfactory attractor ($M = 2.35, SD = 1.14$), and visual attractor ($M = 2.90, SD = 1.29$). When participants heard the dog sound and saw the associated visual feedback, the naturalness and immersion of the auditory attractor increased. The multi-attractor showed a value between those by the auditory and olfactory attractors because the dog sound and floral scent are difficult to associate with one another, thereby decreasing the naturalness and immersion. There is a need to consider the user's imagination-associated stimulation when designing a natural and immersive human-sense-stimulating attractor. All participants experienced the smell for the first time in VR, and the floral scent was an artificial one, which may not feel natural to some participants. Therefore, the naturalness of the olfactory attractor was lower than that of the visual attractor. In terms of immersion, while using the bird object as a visual reorientation trigger was content-relevant, it showed the lowest immersion value. The visual attractor may be considered annoying and could disturb

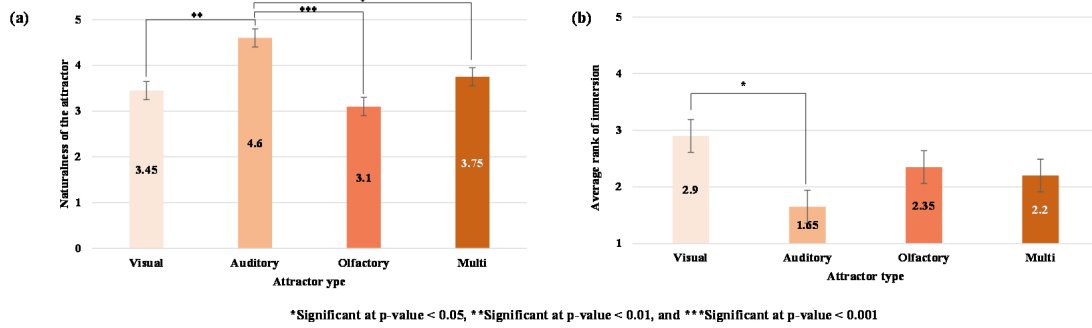


Figure 4: Participants' evaluation of the attractors: (a) naturalness and (b) immersion.

the immersive experience because it constantly requires the walker to look for and follow the attractor.

We obtained participants' opinions on the use of auditory and olfactory attractors to increase their effectiveness and expand their application scope. Six participants stated "I could not find the accurate direction of the olfactory origin", and the direction in which the scent was sprayed did not indicate the direction in which participants had to go. As a human's sense of smell cannot accurately detect the scent's direction, there was no need to express an accurate direction. In contrast, a higher number of direction turn increased the rotation gain, but the frequent rotation and the cognitive load can induce fatigue in the walkers owing to confusion regarding the guiding direction. Therefore, further studies are required to prevent high cognitive load and reduce the sickness. In this experiment, we used one dog sound and one floral scent; however, participants requested a more diverse, content-relevant, and pleasant sound and scent. Diverse sounds and scents are also useful in modulating spatial attention and overcoming auditory and olfactory habituation [2, 16]. Nine participants suggested VR applications of auditory and olfactory attractors are fire drills and finding-activity-based games, such as treasure hunts or escape games, as guides at touring places, for detecting odors that are not easily encountered in daily life, such as in botanical gardens and coasts.

5 CONCLUSION

RDW techniques visually manipulate the VE to change the VR walker's trajectory without noticing the manipulation. The purpose of redirection is to allow the walker to move freely in the tracking area and to prevent accidents such as falls or crashes. An attractor is a rotational RDW method that manipulates the virtual world after capturing the walker's visual engagement. An attractor is effective in keeping the walker in the tracking area when the walker recognizes the attractor. However, the predefined rotational gain and frequent interventions for reorientation can disturb the walker's experience, and the attractor might not suitably reorient the walker without engagement with the walker. To address the limitations of visual attractors, we developed human-sense-stimulating attractors that consist of a reorientation trigger and visual feedback. Taking into consideration the invisibility and direction properties of sound and scent and the inaccuracies of humans' hearing and smell,

we utilized the auditory and olfactory stimulus in our human-sense-stimulating attractors. We compared four different attractor types to test whether the human-sense-stimulating attractors could improve the shortages of the visual attractor. The results showed that the olfactory attractor induced more walker turn changes than the other attractors. The rotational angle is proportional to the walker's movement; therefore, a high number of turn direction changes of the walker could increase the angle of RDW manipulation [5, 22]. For instance, if the clockwise and counter-clockwise rotation behavior of the user is augmented and the rotation degree is accumulated, then the rotational angle can be increased with minimal intervention of the redirection manipulation. A larger RDW angle could decrease the need for intervention by the attractor, and, therefore, decrease the disruption to a walker's experience. The auditory attractor had the highest value of reorientation success rate, naturalness, and immersion among all attractors, demonstrating that it is a promising alternative to overcome the disadvantages of the visual attractor. Without continuous linkage by gaze, auditory and olfactory stimulation can catch the walker's attention and reorient the walker in the desired direction. In future studies, it will be necessary to determine whether promising attractors can actually redirect the walker in a virtual world where the RDW algorithm is applied [11]. The improvement in several factors that decrease the naturalness and immersion scores is also required, for example by designing the scent spraying system without directional spraying or using a more realistic scent or sound. The auditory and olfactory attractors could effectively reorient the walker toward the desired location without disturbing the original purpose in diverse applications.

ACKNOWLEDGMENTS

This work was supported by the GIST-MIT Research Collaboration grant funded by the GIST in 2022.

REFERENCES

- [1] Judith Amores, Robert Richer, Nan Zhao, Pattie Maes, and Bjoern M. Eskofier. 2018. Promoting relaxation using virtual reality, olfactory interfaces and wearable EEG. In 2018 IEEE 15th international conference on wearable and implantable body sensor networks (BSN), 98–101. <https://doi.org/10.1109/BSN.2018.8329668>
- [2] Jennifer C. Brookes. C. 2011. Olfaction: the physics of how smell works?. *Contemporary Physics*, 52(5), 385–402. <https://doi.org/10.1080/00107514.2011.597565>
- [3] Gerd Bruder, Paul Lubos, and Frank Steinicke. 2015. Cognitive resource demands of redirected walking. *IEEE transactions on visualization and computer graphics*,

- 21(4), 539–544. <https://doi.org/10.1109/TVCG.2015.2391864>
- [4] Haiwei Chen and Henry Fuchs. 2017. Supporting free walking in a large virtual environment: imperceptible redirected walking with an immersive distractor. In *Proceedings of the Computer Graphics International Conference*, 1–6. <https://doi.org/10.1145/3095140.3095162>
- [5] Robbe Cools and Adalberto L. Simeone. 2019. Investigating the effect of distractor interactivity for redirected walking in virtual reality. In *Symposium on Spatial User Interaction*, 1–5. <https://doi.org/10.1145/3357251.3357580>
- [6] Dmitrijs Dmitrenko, Emanuela Maggioni, and Marianna Obrist. 2017. OSpace: towards a systematic exploration of olfactory interaction spaces. In *Proceedings of the 2017 ACM International Conference on Interactive Surfaces and Spaces*, 171–180. <https://doi.org/10.1145/3132272.3134121>
- [7] Tobias Feigl, Eliise Köre, Christopher Mutschler, and Michael Philippsen. 2017. Acoustical manipulation for redirected walking. In *Proceedings of the 23rd ACM Symposium on Virtual Reality Software and Technology*, 1–2. <https://doi.org/10.1145/3139131.3141205>
- [8] Camille Ferdenzi, S. Craig Roberts, Annett Schirmer, Sylvain Delplanque, Sezen Kekic, Christelle Porcherot, Isabelle Cayeux, David Sander, and Didier Grandjean. 2013. Variability of affective responses to odors: culture, gender, and olfactory knowledge. *Chemical senses*, 38(2), 175–186. <https://doi.org/10.1093/chemse/bjs083>
- [9] Jann Philipp Freiwald, Oscar Ariza, Omar Janeh, and Frank Steinicke. 2020. Walking by cycling: A novel in-place locomotion user interface for seated virtual reality experiences. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–12. <https://doi.org/10.1145/3313831.3376574>
- [10] Peizhong Gao, Keigo Matsumoto, Takuji Narumi, and Michitaka Hirose. 2020. Visual-Auditory Redirection: Multimodal Integration of Incongruent Visual and Auditory Cues for Redirected Walking. In *2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 639–648. <https://doi.org/10.1109/ISMAR50242.2020.00092>
- [11] Timofey Grechkin, Mahdi Azmandian, Mark Bolas, and Evan Suma. 2015. Towards context-sensitive reorientation for real walking in virtual reality. In *2015 IEEE Virtual Reality (VR)*, 185–186. <https://doi.org/10.1109/VR.2015.7223357>
- [12] Omar Janeh, Eike Langbehn, Frank Steinicke, Gerd Bruder, Alessandro Gulberti, and Monika Poetter-Nerger. 2017. Walking in virtual reality: Effects of manipulated visual self-motion on walking biomechanics. *ACM Transactions on Applied Perception (TAP)*, 14(2), 1–15. <https://doi.org/10.1145/3022731>
- [13] Shenbing Kuang and Tao Zhang. 2014. Smelling directions: olfaction modulates ambiguous visual motion perception. *Scientific reports*, 4(1), 1–5. <https://doi.org/10.1038/srep05796>
- [14] Eike Langbehn, Paul Lubos, Gerd Bruder, and Frank Steinicke. 2017. Application of redirected walking in room-scale VR. In *2017 IEEE Virtual Reality (VR)*, 449–450. <https://doi.org/10.1109/VR.2017.7892373>
- [15] Eike Langbehn, Paul Lubos, Gerd Bruder, and Frank Steinicke. 2017. Bending the curve: Sensitivity to bending of curved paths and application in room-scale vr. *IEEE transactions on visualization and computer graphics*, 23(4), 1389–1398. <https://doi.org/10.1109/TVCG.2017.2657220>
- [16] Rinaldi Luca Maggioni, Emanuela Olivero, Nadia Maravita, Angelo, and Girelli Luisa. 2018. Smelling the space around us: Odor pleasantness shifts visuospatial attention in humans. *Emotion*, 18(7), 971. <https://doi.org/10.1037/emo0000335>
- [17] Emanuela Maggioni, Robert Cobden, Dmitrijs Dmitrenko, Kasper Hornbæk, and Marianna Obrist. 2020. SMELL SPACE: mapping out the olfactory design space for novel interactions. *ACM Transactions on Computer-Human Interaction (TOCHI)*, 27(5), 1–26. <https://doi.org/10.1145/3402449>
- [18] Morgan McCullough, Hong Xu, Joel Michelson, Matthew Jackoski, Wyatt Pease, William Cobb, William Kalescky, Joshua Ladd, and Betsy Williams. 2015. Myo arm: swinging to explore a VE. In *Proceedings of the ACM SIGGRAPH Symposium on Applied Perception*, 107–113. <https://doi.org/10.1145/2804408.2804416>
- [19] Niall Murray, Brian Lee, Yuansong Qiao, and Gabriel-Miro Muntean. 2016. Olfaction-enhanced multimedia: A survey of application domains, displays, and research challenges. *ACM Computing Surveys (CSUR)*, 48(4), 1–34. <https://doi.org/10.1145/2816454>
- [20] Niels Christian Nilsson, Tabitha Peck, Gerd Bruder, Eri Hodgson, Stefania Serafin, Mary Whitton, Frank Steinicke, and Evan Suma Rosenberg. 2018. 15 years of research on redirected walking in immersive virtual environments. *IEEE computer graphics and applications*, 38(2), 44–56. <https://doi.org/10.1109/MCG.2018.11125628>
- [21] Malte Nogalski and Wolfgang Fohl. 2016. Acoustic redirected walking with auditory cues by means of wave field synthesis. In *2016 IEEE Virtual Reality (VR)*, 245–246. <https://doi.org/10.1109/VR.2016.7504745>
- [22] Tabitha C. Peck, Henry Fuchs, and Mary C. Whitton. 2009. Evaluation of reorientation techniques and distractors for walking in large virtual environments. *IEEE transactions on visualization and computer graphics*, 15(3), 383–394. <https://doi.org/10.1109/TVCG.2008.191>
- [23] Tabitha C. Peck, Henry Fuchs, and Mary C. Whitton. 2011. An evaluation of navigational ability comparing Redirected Free Exploration with Distractors to Walking-in-Place and joystick locomotion interfaces. In *2011 IEEE Virtual Reality Conference*, 55–62. <https://doi.org/10.1109/VR.2011.5759437>
- [24] Tabitha C. Peck, Henry Fuchs, and Mary C. Whitton. 2011. The design and evaluation of a large-scale real-walking locomotion interface. *IEEE transactions on visualization and computer graphics*, 18(7), 1053–1067. <https://doi.org/10.1109/TVCG.2011.289>
- [25] Jess Porter, Brent Craven, Rehan M Khan, Shao-Ju Chang, Irene Kang, Benjamin Judkewitz, Jason Volpe, Gary Settles and Noam Sobel. 2007. Mechanisms of scent-tracking in humans. *Nature neuroscience*, 10(1), 27–29. <https://doi.org/10.1038/nn1819>
- [26] Michael I. Posner, Mary Jo Nissen, and Raymond M. Klein. 1976. Visual dominance: an information-processing account of its origins and significance. *Psychological review*, 83(2), 157. <https://doi.org/10.1037/0033-295X.83.2.157>
- [27] Nimesha Ranasinghe, Koon Chuan Raymond Koh, Nguyen Thi Ngoc Tram, Yan Liangkun, Kala Shamaiah, Siew Geuk Choo, David Tolley, Shienny Karwita, Barry Chew, Daniel Chua, and Ellen Yi-Luen Do. 2019. Tainted: An olfaction-enhanced game narrative for smelling virtual ghosts. *International Journal of Human-Computer Studies*, 125, 7–18. <https://doi.org/10.1016/j.ijhcs.2018.11.011>
- [28] Nimesha Ranasinghe, Pravar Jain, Nguyen Thi Ngoc Tram, Koon Chuan Raymond Koh, David Tolley, Shienny Karwita, Lin Lien-Ya, Yan Liangkun, Kala Shamaiah, Chow Eason Wai Tung, Ching Chuan Yen, and Ellen Yi-Luen Do. 2018. Season traveller: Multisensory narration for enhancing the virtual reality experience. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–13. <https://doi.org/10.1145/3173574.3174151>
- [29] Sharif Razzaque, Zachariah Kohn, and Mary C. Whitton. 2001. Redirected walking. In *Proceedings of EUROGRAPHICS*, 9, 105–106. Manchester, UK.
- [30] Nicholas Rewkowski, Atul Rungta, Mary Whitton, and Ming Lin. 2019. Evaluating the effectiveness of redirected walking with auditory distractors for navigation in virtual environments. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 395–404. <https://doi.org/10.1109/VR.2019.8798286>
- [31] Stefania Serafin, Niels C. Nilsson, Erik Sikstrom, Amalia De Goetzen, and Rolf Nordahl. 2013. Estimation of detection thresholds for acoustic based redirected walking techniques. In *2013 IEEE Virtual Reality (VR)*, 161–162. <https://doi.org/10.1109/VR.2013.6549412>
- [32] Misha Sra, Xuhai Xu, Aske Mottelson, and Pattie Maes. 2018. VMotion: designing a seamless walking experience in VR. In *Proceedings of the 2018 Designing Interactive Systems Conference*, 59–70. <https://doi.org/10.1145/3196709.3196792>
- [33] Frank Steinicke, Gerd Bruder, Jason Jerald, Harald Frenz, and Markus Lappe. 2009. Estimation of detection thresholds for redirected walking techniques. *IEEE transactions on visualization and computer graphics*, 16(1), 17–27. <https://doi.org/10.1109/TVCG.2009.62>
- [34] Evan A. Suma, Gerd Bruder, Frank Steinicke, David M. Krum, and Mark Bolas. 2012. A taxonomy for deploying redirection techniques in immersive virtual environments. In *2012 IEEE Virtual Reality Workshops (VRW)*, 43–46. <https://doi.org/10.1109/VR.2012.6180877>
- [35] Evan A. Suma, Seth Clark, David Krum, Samantha Finkelstein, Mark Bolas, and Zachary Warte. 2011. Leveraging change blindness for redirection in virtual environments. In *2011 IEEE Virtual Reality Conference*, 159–166. <https://doi.org/10.1109/VR.2011.5759455>
- [36] Qi Sun, Anjul Patney, Li-Yi Wei, Omer Shapira, Jingwan Lu, Paul Asente, Suwen Zhu, Morgan McGuire, David Luebke, and Arie Kaufman. 2018. Towards virtual reality infinite walking: dynamic saccadic redirection. *ACM Transactions on Graphics (TOG)*, 37(4), 1–13. <https://doi.org/10.1145/3197517.3201294>
- [37] Betsy Williams, Stephen Bailey, Gayathri Narasimham, Muqun Li, and Bobby Bodenheimer. 2011. Evaluation of walking in place on a Wii balance board to explore a virtual environment. *ACM Transactions on Applied Perception (TAP)*, 8(3), 1–14. <https://doi.org/10.1145/2010325.2010329>
- [38] Preston Tunnell Wilson, William Kalescky, Ansel MacLaughlin, and Betsy Williams. 2016. VR locomotion: walking in place to arm swinging. In *Proceedings of the 15th ACM SIGGRAPH Conference on Virtual-Reality Continuum and Its Applications in Industry-Volume 1*, 243–249. <https://doi.org/10.1145/3013971.3014010>